EXCELLENCE

CLASS 11 - 2026-27

Physics | Lecture - 16

Basic Maths and Calculus

Definite Integrals and Area under a Curve

By - Excellence Programme

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Topics to be covered

01 5-minute indefinite-integral recap

02 Definite integral âˆ«_a^b f(x) dx

03 Fundamental rule: F(b) âˆ’ F(a)

04 No +C in definite integrals

05 Area under a curve

06 Easy definite-integral practice

07 Physics: displacement = area under vâ€“t

Indefinite integrals from last class

âˆ« f(x) dx = F(x) + C

âˆ« xâ¿ dx = xâ¿âºÂ¹/(n+1) + C (n â‰ â€“1)

âˆ« sin x dx = â€“cos x + C Â· âˆ« cos x dx = sin x + C

Check by differentiating: you must recover f(x)

Quick check: indefinite

Find âˆ« (3xÂ² â€’ 4) dx.

Solution: Quick check: indefinite

Step 1: âˆ« 3xÂ² dx = xÂ³

Step 2: âˆ« âˆ’4 dx = âˆ’4x

xÂ³ âˆ’ 4x + C.

Rough work / extra work

Definite integral

A definite integral has lower and upper limits. The answer is a number.

$$\hat{a}\hat{}\text{«}_a\hat{}b\ f(x)\ dx = [F(x)]_a\hat{}b = F(b)\ \hat{a}\hat{}\text{'}\ F(a)$$

F is any antiderivative of f. The constant C cancels, so we omit +C.

Never write +C in a definite integral.

Three easy steps

1. Find the antiderivative F(x) of f(x)

2. Put the upper limit: F(b)

3. Subtract the lower limit: F(b) â^’ F(a)

Answer = F(upper) â^’ F(lower)

First definite integral

Evaluate âˆ«_0Â² 2x dx.

Solution: First definite integral

Step 1: Antiderivative: F(x) = xÂ²

Step 2: F(2) âˆ’ F(0) = 4 âˆ’ 0

âˆ«_0Â² 2x dx = 4.

Power definite integral

Evaluate âˆ«_1Â³ xÂ² dx.

Solution: Power definite integral

Step 1: $F(x) = xÂ^3/3$

Step 2: $F(3) â^' F(1) = 27/3 â^' 1/3 = 9 â^' 1/3$

26/3.

Rough work / extra work

Polynomial definite

Evaluate âˆ«_0^1 (4xÂ³ âˆ’ 2x) dx.

Solution: Polynomial definite

Step 1: F(x) = xâ①´ â^' xÂ²

Step 2: F(1) â^' F(0) = (1 â^' 1) â^' 0

0.

Trig definite

Evaluate âˆ«_0^{Ï€/2} cos x dx.

Solution: Trig definite

Step 1: $F(x) = \sin x$

Step 2: $F(\pi/2) - F(0) = \sin(\pi/2) - \sin 0 = 1 - 0$

1.

Rough work / extra work

Sine definite

Evaluate âˆ«_0^{Ï€/2} sin x dx.

Solution: Sine definite

Step 1: F(x) = â^’cos x

Step 2: F(Ï€/2) â^’ F(0) = (â^’cos Ï€/2) â^’ (â^’cos 0) = 0 â^’ (â^’1)

1.

Area under a curve

When f(x) â‰¥ 0 on [a, b], the definite integral equals the area between the curve and the x-axis.

Area = âˆ«_a^b f(x) dx (f â‰¥ 0)

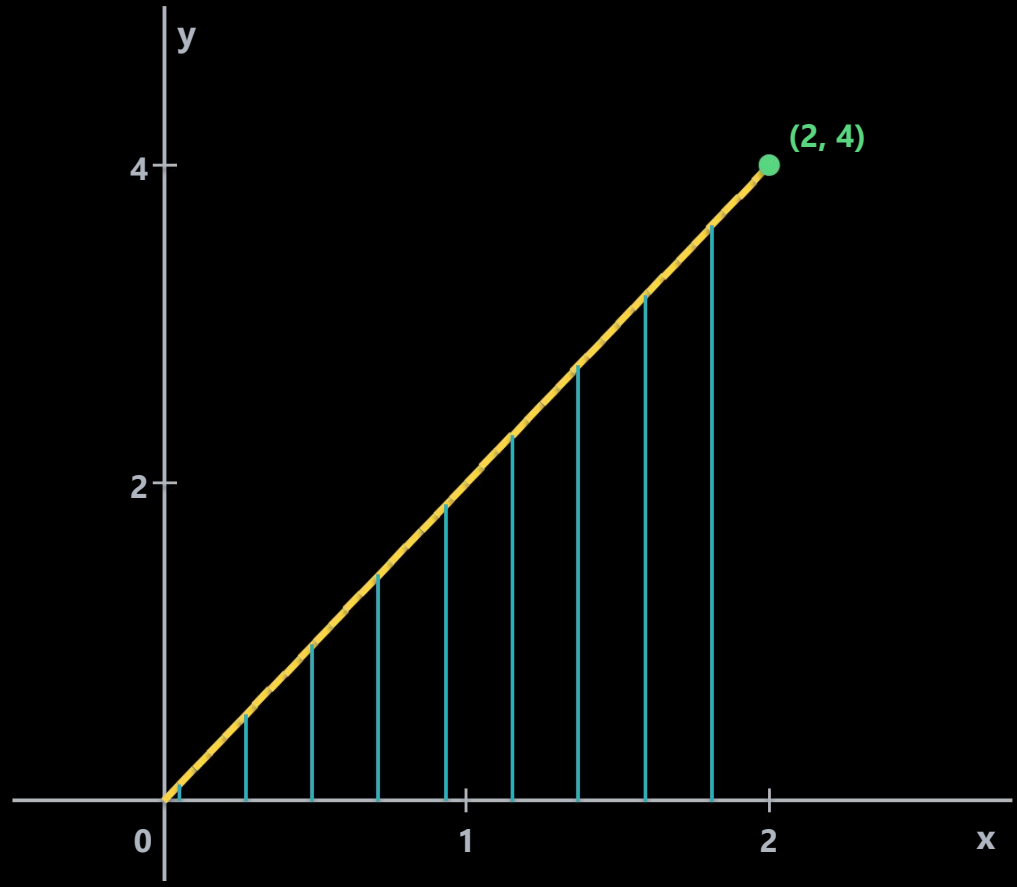
If the curve dips below the axis, the integral counts that region as negative.

Area of a region below the axis needs absolute value (or split the integral).

Area under y = 2x from 0 to 2

The region is a right triangle of base 2 and height 4.

$$\text{Area} = \tfrac{1}{2} \cdot 2 \cdot 4 = 4 = \int_0^2 2x\,dx$$



Area under a line

Find the area under $y = x$ from $x = 0$ to $x = 4$.

Solution: Area under a line

Step 1: Area = âˆ«_0â①´ x dx = [xÂ²/2]_0â①´

Step 2: = 16/2 â^' 0 = 8

Area = 8 square units.

Rough work / extra work

Area under a parabola

Find the area under $y = xÂ^2$ from $x = 0$ to $x = 2$.

Solution: Area under a parabola

Step 1: Area = âˆ«_0Â² xÂ² dx = [xÂ³/3]_0Â²

Step 2: = 8/3 â€“ 0

Area = 8/3 square units.

Area under cosine

Find the area under y = cos x from x = 0 to x = Ï€/2.

Solution: Area under cosine

Step 1: Area = âˆ«_0^{Ï€/2} cos x dx = [sin x]_0^{Ï€/2}

Step 2: = 1 â€“ 0 = 1

Area = 1 square unit.

Rough work / extra work

Two useful properties

$$\hat{a}^{\wedge}\!\!\ll_a^b f = \hat{a}^{\wedge}\!\!\prime\ \hat{a}^{\wedge}\!\!\ll_b^a f$$

$$\hat{a}^{\wedge}\!\!\ll_a^b f = \hat{a}^{\wedge}\!\!\ll_a^c f + \hat{a}^{\wedge}\!\!\ll_c^b f$$

You may split an interval at a convenient point c.

Swapping limits changes the sign of the integral.

Swap limits

If âˆ«_0Â² 2x dx = 4, find âˆ«_2â① ° 2x dx.

Solution: Swap limits

Step 1: Swapping limits multiplies by âˆ’1

âˆ«_2â①° 2x dx = âˆ’4.

Split the interval

Evaluate âˆ«_0â① x dx by splitting at x = 2.

Solution: Split the interval

Step 1: âˆ«_0â①´ = âˆ«_0Â² + âˆ«_2â①´

Step 2: âˆ«_0Â² x dx = 2 Â· âˆ«_2â①´ x dx = [xÂ²/2]_2â①´ = 8 â^’ 2 = 6

2 + 6 = 8.

Rough work / extra work

Constant integrand

Evaluate âˆ«_1â① µ 3 dx.

Solution: Constant integrand

Step 1: $F(x) = 3x$

Step 2: $F(5) - F(1) = 15 - 3 = 12$

12.

Exponential definite

Evaluate âˆ«_0^1 eË£ dx.

Solution: Exponential definite

Step 1: F(x) = eË£

Step 2: F(1) â^’ F(0) = e â^’ 1

e â^’ 1.

Mixed polynomial

Evaluate âˆ«_0Â² (xÂ³ + 2x) dx.

Solution: Mixed polynomial

Step 1: $F(x) = xâ①´/4 + xÂ^{2}$

Step 2: $F(2)$ âˆ’ $F(0) = (4 + 4)$ âˆ’ $0 = 8$

8.

Rough work / extra work

Physics: area under vâ€“t graph

On a velocityâ€“time graph, the area between the curve and the t-axis is displacement.

$$\text{displacement} = \hat{a}\hat{}\text{«_\{t\hat{a},①\}\^{}\{t\hat{a},,\}}\ v(t)\,dt = \text{area under v\hat{a}€“t}$$

Same idea as last class: integrating velocity recovers position change.

Area below the t-axis means displacement in the opposite direction.

Constant velocity displacement

If v = 5 m sâ①»Â^{1} from t = 0 to t = 4 s, find the displacement.

Solution: Constant velocity displacement

Step 1: $s = \int_0^4 5\,dt = [5t]_0^4 = 20 - 0$

Step 2: Area of rectangle = $5 \times 4 = 20$

Displacement = 20 m.

Linear velocity displacement

If v = 2t (m sâ①»Â^{1}) from t = 0 to t = 3 s, find the displacement.

Solution: Linear velocity displacement

Step 1: $s = \hat{a}\hat{}\text{«}_0\hat{A}^3\ 2t\ dt = [t\hat{A}^2]_0\hat{A}^3 = 9$

Step 2: Triangle area = $\hat{A}\frac{1}{2}\ \hat{A}\cdot\ 3\ \hat{A}\cdot\ 6 = 9$

Displacement = 9 m.

Rough work / extra work

From a to Î”v

If a = 3 m sâ①»Â² (constant) from t = 0 to t = 2 s and u = 0, find v at t = 2.

Solution: From a to Î”v

Step 1: $\Delta v = \int_0^2 a\,dt = \int_0^2 3\,dt = 6$

Step 2: $v = u + \Delta v = 0 + 6$

$v = 6$ m sâ①»Â¹.

Homework: definite power

Evaluate âˆ«_1Â² (3xÂ² âˆ’ 1) dx.

Solution: Homework: definite power

Step 1: $F(x) = xÂ^3 â^' x$

Step 2: $F(2) â^' F(1) = (8 â^' 2) â^' (1 â^' 1) = 6$

6.

Rough work / extra work

Homework: area under line

Find the area under $y = 3x$ from $x = 1$ to $x = 3$.

Solution: Homework: area under line

Step 1: Area = âˆ«_1Â³ 3x dx = [3xÂ²/2]_1Â³

Step 2: = (27/2) â€“ (3/2) = 24/2 = 12

Area = 12 square units.

Homework: check by graph

v falls from 10 m/s to 0 in 5 s on a straight vâ€“t graph. Find displacement.

Solution: Homework: check by graph

Step 1: Triangle area = Â½ Â· 5 Â· 10 = 25

Step 2: Or: v = 10 â^’ 2t; â^«_0â①µ (10 â^’ 2t) dt = [10t â^’ tÂ²]_0â①µ = 50 â^’ 25

Displacement = 25 m.

Definite-integral lock lines

âˆ«_a^b f(x) dx = F(b) âˆ’ F(a) (no +C)

If f â‰¥ 0, this equals area under the curve from a to b

âˆ«_a^b f = âˆ’âˆ«_b^a f Â· split intervals when useful

Displacement = area under the vâ€“t graph

Rough work / extra work

Practice

1. Redo every definite-integral question without the solution.

2. For each area question, also sketch the region.

3. Convert one vâ€“t triangle or rectangle into an âˆ« statement.

4. Write: indefinite needs +C; definite does not.

Forgetting to subtract F(a) is the most common definite-integral error.